ENEE12015 - Electrical Power Engineering

Lecture Notes

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1 Introduction to Electrical Power Systems

1.1 Early Development:

1800's - Electricity and Magnetism well understood intellectually
1831 - Magnetica field was discovered to generate electricity
1845 - First commerical operations of telegraph
1860 - First electrical generator
1870 - First incandesent light bulb
1882 - First commericial power system in NY City
1883 - Polyphase induction motor
1885 - Commericial availability of AC power

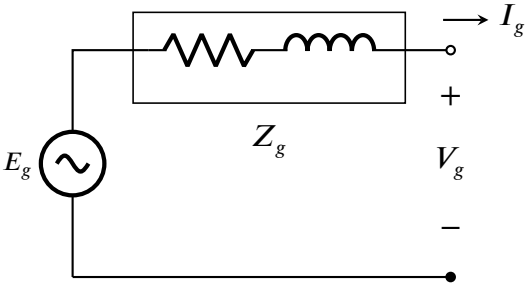
1.2 Power System Compenets:

- Generators
- Transformers
- Transmission lines
- Substations/Terminal Stations
- Distribution lines
- Electrical loads

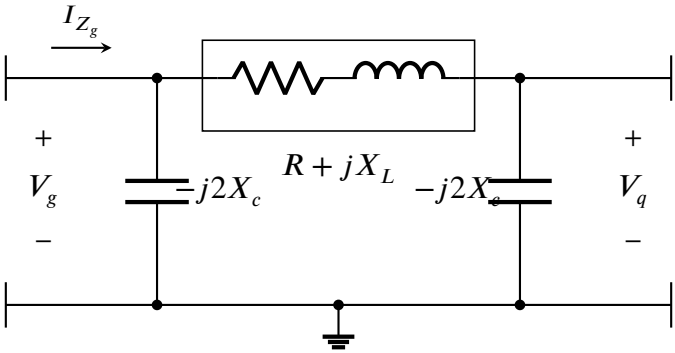
When transmitting power, the best senario is high voltage, low current transmission. Ideally below the resistivity of the power lines.

1.3 Power System Models:

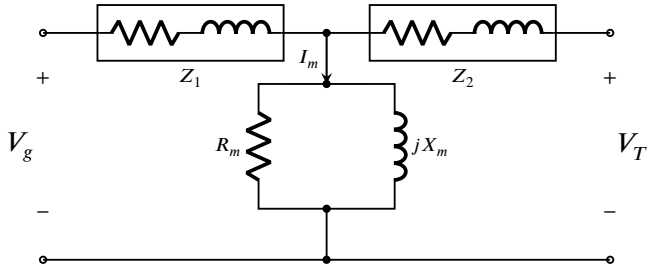
1.3.1 Generators



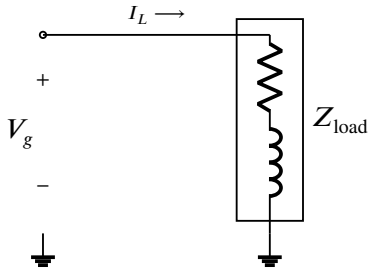
1.3.2 Transmission/Distribution Lines



1.3.3 Transformers



1.3.4 Loads



1.4 Basic Theory:

Ohm's Law - $V = IZ$
KVL - $\sum V = 0$

KCL - $\sum I_{in} = \sum I_{out}$
Power Flow - $S = P + jQ = VI^*$

1.5 Current and Power Flow:

- Technically current flow within an AC system is arbitrary
- Power is clear from source to load
- Steady state is only assumed

1.6 Power in AC Circuits:

Average Power over time period T

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)dt$$

Instantaneous Power

$$p(t) = v(t)i(t)$$

When a circuit is in steady state, all voltages and currents become sinusodial. The periodic nature of the waveforms enables more specific derivation of power relationships.

$$P_{av} = \frac{E_m I_m}{2} = \frac{E_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} = E_{RMS} I_{RMS}$$

This only works when $v(t)$ and $i(t)$ are in phase. When a small phase shift occurs:

$$P_{av} = \frac{E_m I_m}{2}\cos(\theta) = E_{RMS} I_{RMS}\cos(\theta)$$

When $i(t)$ lags $v(t)$ by 90°

$$P_{av} = E_{RMS} I_{RMS}\cos(90°) = 0$$

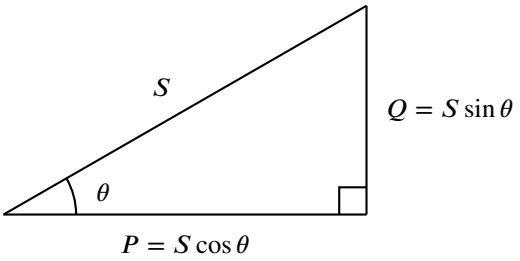
1.7 Reactive Power:

Power that dose no useful work within a system.

$$Q = E_{RMS} I_{RMS}\sin(\phi))$$

VAR is a defined value and does not relate to the area under P. When $v(t0$ and $i(t)$ are in phase ($\phi = 0$), $Q = 0$. When reactive power is maximum ($\phi = 90°$), $Q = E_{RMS} I_{RMS}$ and $P_{av} = 0$.

1.8 Power Triangles:



Q is wished to be minised, allowing the maximum S to used within the load (active power) usefully.

1.9 Power Flow Direction:

A positive P present in a power triangle indicates that ϕ is between $-90°$ and $90°$. With a negative P show ϕ to be $> 90°$. It is necessary to assign a current direction in AC circuits, for the purpose of writing equations in a consistent manner. $Q > 0$, flowing in the same direction as current is assumed to go. $Q < 0$, flowing apposed to the assumed current direction.
Power Factor:

$$pf = \frac{P}{S} = \cos(\phi)$$

2 Three Phase Power

2.1 What is it?:

A 3-phase power system is just a connecting of 3 singlephase systems together for set operation. Each phase must operate at the same voltage and current magnitudes seperated by $120°$ phase angle.

2.2 Balanced Operating Conditions:

Under normal operating conditions all voltages, current, $120°$ angle shift and loads contain same characteristics.

2.3 Analysis of 3-Phase:

Assuming a system is under balanced conditions, means that only a single phase system needs to be analysed. Faults, abnormal load conditions, cause a system

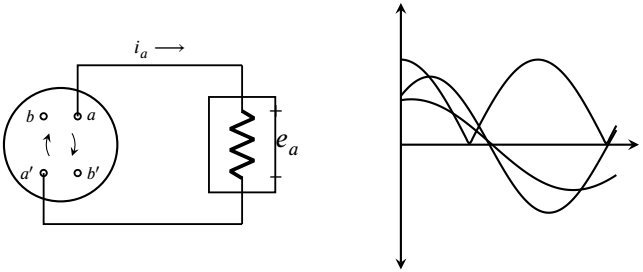
to become unbalanced.

2.4 Popularity of Polyphase Systems:

Polyphase machines can be designed to operate under constant total power conditions. Hence, these machines have smoother operations. These machines are then simpler to design, how all as tansmission lines can delivers greater power for less cost.

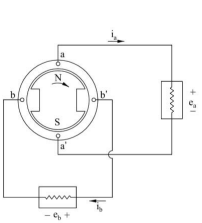
2.5 Power in Polyphase Systems:

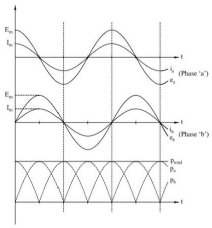
Single Phase: Resistive



$$
\begin{aligned}
p_a &= e_a i_a \\
&= E_m I_m \cos^2(\omega t) \\
P_{av} &= E_{RMS} I_{RMS}
\end{aligned}
$$

2-Phase: Resistive





Voltage induced in b-b' lags voltage in a-a' by 90 electrical degrees. In 2-pole machines, 1 electrical degree is equivalent to 1 mechanical degree.

$$\begin{aligned} e_a &= E_m \cos(\omega t) \\ i_a &= I_m \cos(\omega t) \end{aligned}$$

$$\begin{aligned} e_b &= E_m \cos(\omega t - 90°) \\ &= E_m \sin(\omega t) \\ i_b &= I_m \cos(\omega t - 90°) \\ &= I_m \sin(\omega t) \end{aligned}$$

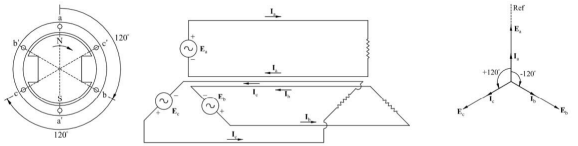
$$\begin{aligned} P_{Tot} &= P_a + P_b \\ &= E_m I_m \cos^2(\omega t) + E_m I_m \sin^2(\omega t) \\ &= E_m I_m \end{aligned}$$

There are two seperate loads. This P_{Tot} is true as long as the load is the same.

$$\begin{aligned} P_{av} &= P_{Tot} \\ &= 2E_{RMS} I_{RMS} \end{aligned}$$

$$\begin{aligned} S_{Tot} &= S_a + S_b \\ &= E_a I_a^* + E_b I_b^* \\ &= 2EI\angle 0° \end{aligned}$$

3-Phase: Resistive



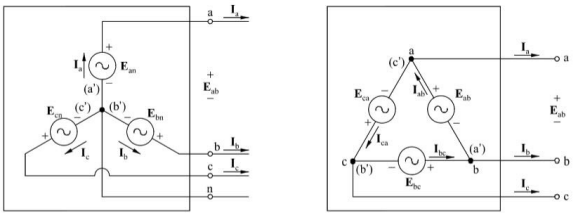
$$
\begin{aligned}
e_a &= E_m \cos(\omega t) \\
i_a &= I_m \cos(\omega t)
\end{aligned}
$$

$$
\begin{aligned}
e_b &= E_m \cos(\omega t - 120°) \\
i_b &= I_m \cos(\omega t - 120°)
\end{aligned}
$$

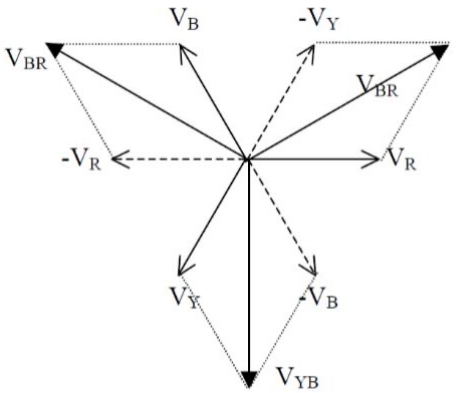
$$
\begin{aligned}
e_c &= E_m \cos(\omega t - 240°) \\
&= E_m \cos(\omega t + 120°) \\
i_c &= I_m \cos(\omega t - 240°) \\
&= I_m \cos(\omega t + 120°)
\end{aligned}
$$

$$
\begin{aligned}
P_{Tot} &= P_a + P_b + P_c \\
&= 1.5 E_m I_m \\
&= 3 E_{RMS} I_{RMS}
\end{aligned}
$$

2.6 3-Phase Connections:



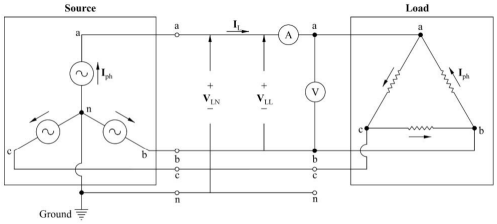
2.7 Balanced Systems:



- Angles between phases is 120°
- Magnitudes are all equal
- V_L would be 30° leading nearest phase voltage
- $|V_L|$ is $\sqrt{3}|V_P|$

$$I_L = I_P$$
$$|V_L| = \sqrt{3}|V_P|$$
$$|I_L| = \sqrt{3}|I_P|$$

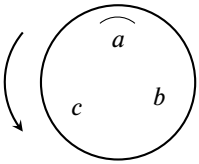
2.8 3-Phase Device Connection:



The delta-star and star-delta conversions can be done. The only requirement is that the device voltage ratings must be compatitable.

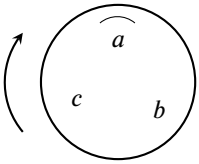
2.9 Phase Sequence:

The phase sequence denotes which phase becomes succesively positive.



+ve seq

$-a-b-c-a-b-c-$



–ve seq

$-a-c-b-a-c-b-$

2.10 3-Phase Solutions:

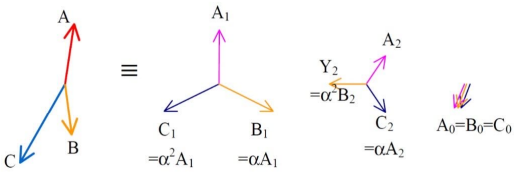
- Most 3-Phase system can be though of as balanced
- To solve an unbalanced 3-phase system is quite complex

2.11 Symmetrical Component Analysis:

Any unbalanced system can be split up into a series of balanced systems.

- Same phase sequence (Positive)
- Opposite phase sequence (Negative)
- In phase quantities (Zero)

2.12 Unbalanced Decomposition:

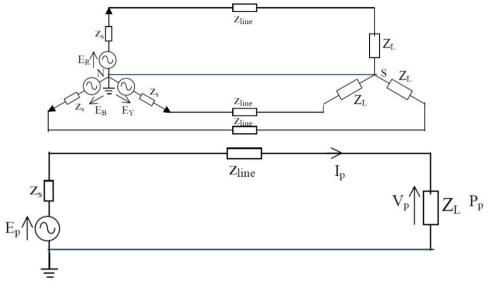


$$\alpha = 1\angle 120^\circ$$

$$A = A_0 + A_1 + A_2 = A_0 + A_1 + A_2$$
$$B = B_0 + B_1 + B_2 = A_0 + \alpha^2 A_1 + \alpha A_2$$
$$C = C_0 + C_1 + C_2 = A_0 + \alpha A_1 + \alpha^2 A_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix}$$

2.13 Balanced Analysis:



2.13.1 Wye Connection

$$I_P = I_L$$
$$V_P = \frac{V_L}{\sqrt{3}}$$

2.13.2 Delta Connection

$$V_P = V_L$$
$$I_P = \frac{I_L}{\sqrt{3}}$$

2.13.3 All

$$S_{3P} = 3V_P I_P = \sqrt{3} V_L I_L$$
$$P_{1P} = \frac{P_{3P}}{3}$$

2.14 Selecting Number of Phases:

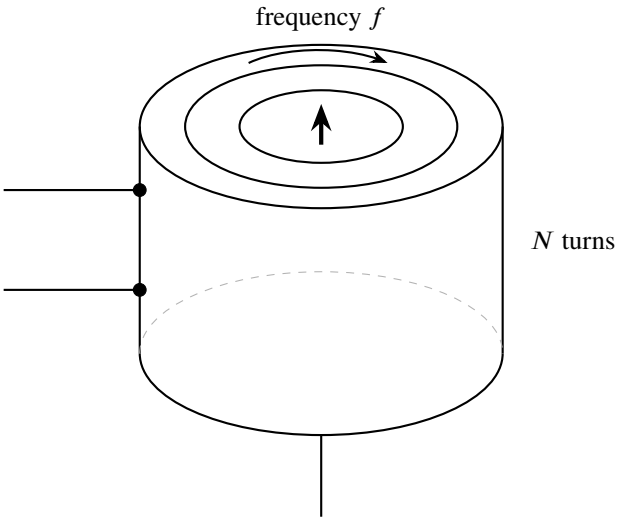
The number of phases for a power system is determined by transmission system economies. That is done by analyzing the amount of power transferred per conductor. With the assumption that the size of the return conductor is equal to that phase conductor.

System	Power Rating	No. Conductors	Power per Conductor
1-Phase	P	2	0.5P
2-Phase	2P	3	0.667P
3-Phase	3P	4	0.75P
4-Phase	4P	5	0.8P

3 Transformers

3.1 Voltage Induced in a Coil:

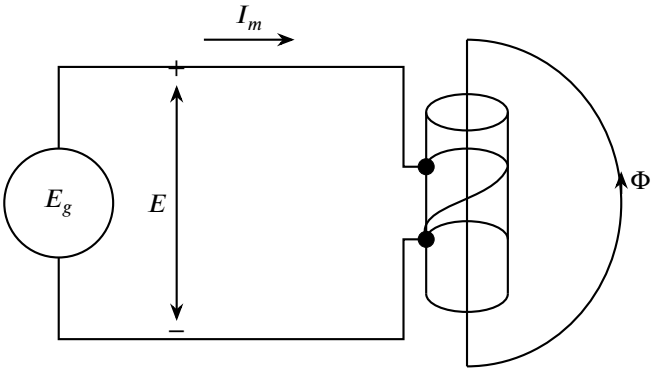
A voltage is induced in a coil when it links to a variable flux. The variable flux then induces a variable voltage.



$$E = 4.44 f N \phi_{max}, \text{ Where:}$$

- E - Effective induced voltage (V)
- f - Frequency (Hz)
- N - Number of turns within the coil
- ϕ_{max} - Peak flux value (Wb)

3.2 Using Voltage:



If the coil resistence is negligible, reactance is:

$$X_m = \frac{E_g}{I_m}$$

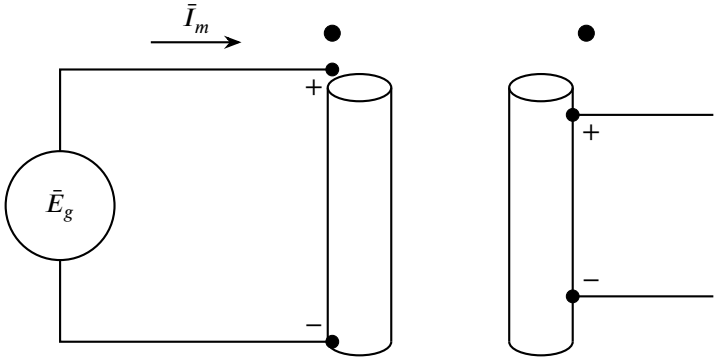
$$\phi_{max} = \frac{E_g}{4.44 f N}$$

3.3 Core Types:

The most common are air and iron. For the same E_g, the ϕ_{max} will be the same. The only difference is that the iron core requires less current.

3.4 Elementray Transformers:

The ϕ_{max1} can induce voltage E_{g2}. This is done throuhgh multual (ϕ_{m1} and leakage (ϕ_{f1}) flux. Dot notation is used to know the polarity of the induced voltage.



A current entering a polarity marked terminal produces a flux in a positive direction.

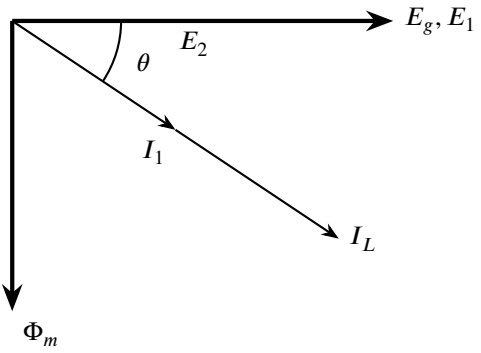
3.5 Ideal Transformer:

- An ideal transformer has no losses in its core, as it is infinitely permeable
- Fluxes are completely mutual with no leakage
- Polarity induced depends on the winding direction relative to the primary

$$E_1 = 4.44 f N_1 \phi_{max}$$
$$E_2 = 4.44 f N_2 \phi_{max}$$
$$a = \frac{E_1}{E_2} = \frac{N_1}{N_2}$$

3.6 Ideal Under Load:

- The mutual flux remains unchanged

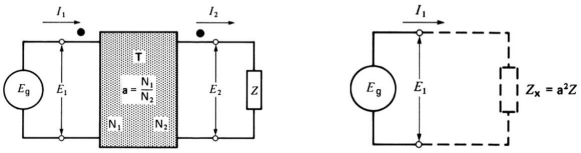


- ϕ_m can remain the same only if the primary side develops an mmf, which is counter-balanced by the secondary mmf, at all times

$$N_1 I_1 = N_2 I_2$$
$$a = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

3.7 Impedance Transformation:

3.7.1 Secondary to Primary

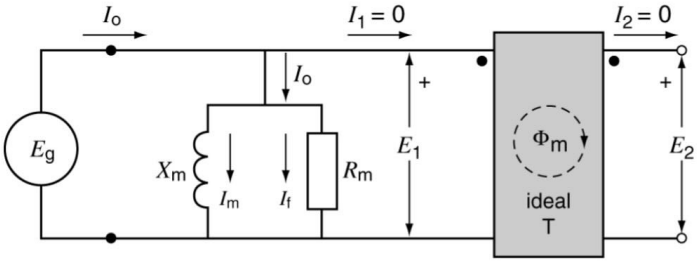


The effective impedance (Z_x) seen by E_g is:

$$
\begin{aligned}
Z_x &= \frac{E_1}{I_1} \\
Z &= \frac{E_2}{I_2} \\
\therefore Z_x &= \frac{E_1}{I_1} = \frac{aE_2}{\frac{I_2}{a}} \\
&= \frac{a^2 E_2}{I_2} \\
&= a^2 Z
\end{aligned}
$$

3.8 Actual Transformer Core:

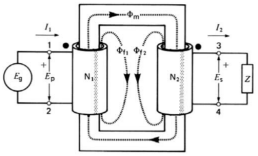
- Not infinitely permeable
- Has hysteresis loss
- Has eddy current loss
  - R_m - Iron core loss
  - X_m - Magnetizing reactance
  - Measure of permeability of transformer core
- R_m draws a small current I_f
- X_m current is I_m
- $E_g = E_1$, I_m lags E_1 by 90°



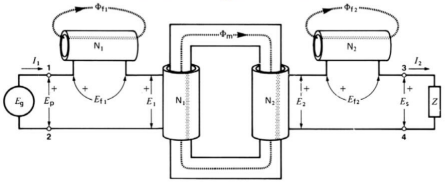
$$R_m = \frac{E_1^2}{P_m}$$

$$X_m = \frac{E_1^2}{Q_m}$$

3.9 Leakage Flux:

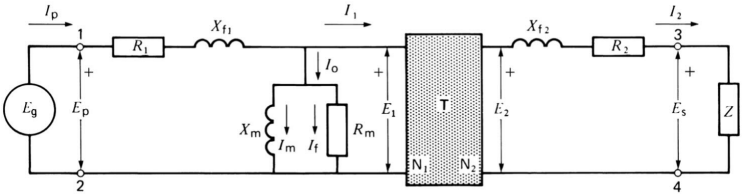


- Separating the various induced voltages due to the mutual flux and the leakage fluxes



Leakage flux occurs due to an imperfect coupling between coils.

3.10 Complete Equivalent Circuit:

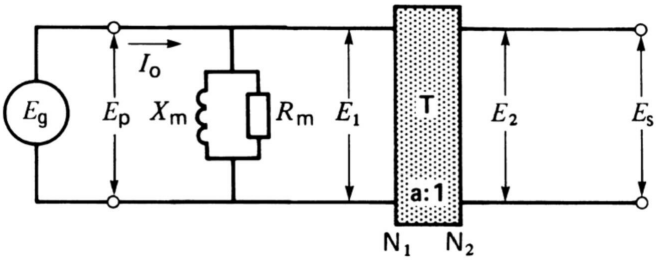


3.11 Simplifing Circuit:

3.11.1 No Load

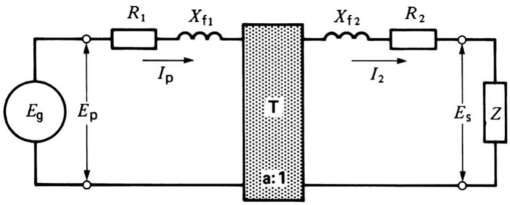
- $I_2 = 0$, as such current $I_1 = 0$

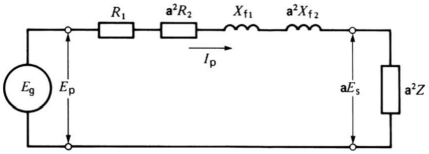
- Since, only I_o flows through R and X_{f1}, voltage drops across them can be neglected
- Turns raio can be found from the ratio of primary voltage and O/C secondary voltage



3.11.2 Full Load

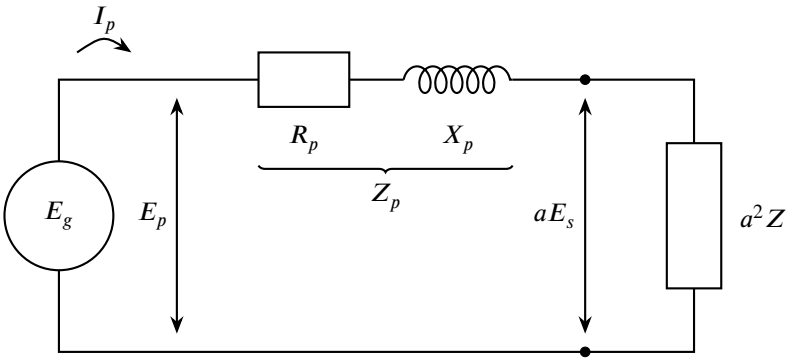
- I_p will be approximately 20 time I_o
- Hence, I_o can be ignored





$$R_p = R_1 + a^2 R_2$$

$$X_p = X_{f1} + a^2 X_{f2}$$



3.12 Measuring Transformer Impedances:

O/C tests and determination of R_m, X_m and a within the core.

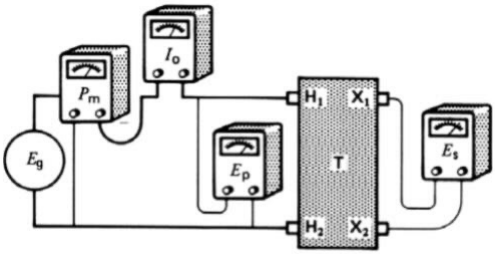
$$S_m = E_p I_o$$

$$Q_m = \sqrt{{S_m}^2 - {P_m}^2}$$

$$R_m = \frac{{E_p}^2}{P_m}$$

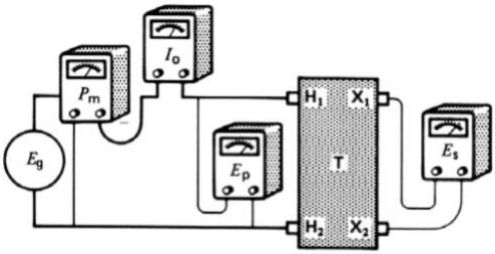
$$X_m = \frac{{E_p}^2}{Q_m}$$

$$a = \frac{E_p}{E_s}$$



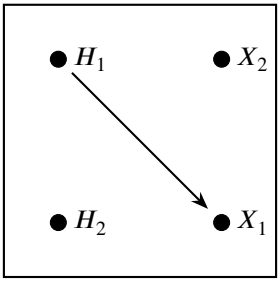
S/C test determines leakage reactance and winding resistance. The applied voltage for S/C testing must be <5% of rated.

$$Z_p = \frac{E_{SC}}{I_{SC}}$$
$$R_p = \frac{P_{SC}}{{I_{SC}}^2}$$
$$X_p = \sqrt{{X_p}^2 - {R_p}^2}$$

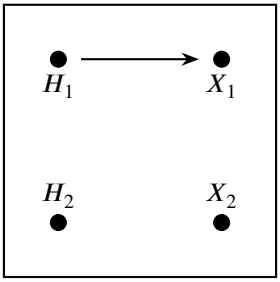


3.13 Terminal Windings:

- H_1, H_2 for the high voltage winding
- X_1, X_2 is the low voltage winding



Additive Polarity



Subtractive Polarity

Subtractive polarity is standard for all single-phase transformers above 200kVA provided the HV winding is 8660V. Most others are additive polarity.